

# Sumcheck-free hypercube inner products from table-form Kronecker encodings

Research note (companion to Kronecker-FRI)

September 2026 — version 9 (Section 5 formalised in Lean; combined lookups)

## Abstract

We encode a multilinear polynomial by its *table*,  $V_f(X) = \sum_w f(w)X^w$ , instead of its coefficients. Two quantities then become single coefficients of univariate products: the evaluation  $\tilde{f}(z)$ , with a product-form kernel, and the hypercube inner product  $\sum_w a(w)b(w)$  of two committed tables, through the reversal  $V_b^*(X) = X^{N-1}V_b(1/X)$ . Because a smooth domain is closed under inversion, the verifier reads  $V_b^*$  from the commitment to  $b$  as a virtual word, and the Kronecker-FRI machinery (opening polynomial, virtual word, one folding test) proves  $\sum_w a(w)b(w) = S$  with no sumcheck, with soundness error  $3M/|\mathbb{F}| + \varepsilon_{\text{fold}} + (1 - \delta)^\kappa$ . With geometric weights  $\gamma^w$  and DEEP quotients as further virtual words, the same method gives a sumcheck-free Hadamard check  $a \circ b = c$  for three committed tables, with soundness error  $2(N - 1)/|\mathbb{F}| + 8M/|\mathbb{F}| + \varepsilon_{\text{fold}} + (1 - \delta)^\kappa$ . Finally, any list of evaluations, inner products and Hadamard checks on the same commitments is proved by one folding test, with soundness error  $2(N - 1)/|\mathbb{F}| + tM/|\mathbb{F}| + \varepsilon_{\text{fold}} + (1 - \delta)^\kappa$  for  $t + 1$  batched words; the term  $2(N - 1)/|\mathbb{F}|$  is paid once, whatever the number of Hadamard checks. The results of Sections 1–4 are machine-checked in Lean 4. Section 5 reduces sparse matrix–vector products, and hence R1CS, to the same statements with lookups by logarithmic derivatives, so an R1CS argument needs no sumcheck at all; its LogUp lemma and lincheck reduction are machine-checked as well.

**Setting.** Notation and parameters are those of Kronecker-FRI (KF) [1]:  $n \geq 1$ ,  $N = 2^n$ , a smooth domain  $L \leq \mathbb{F}^\times$  of order  $M = 2^{n+R}$  with  $R \geq 2$ , codes  $\mathcal{C}_j$ , the folding test  $\Pi_{\text{FT}}[\ell, \kappa]$  with error  $\varepsilon_{\text{fold}}$  (KF Thm 5.24), fibre distance  $\Delta_0^{\text{fib}}$ , and  $0 < \delta \leq \delta^* = (1 - \rho)/2$ . A bit vector  $w \in \{0, 1\}^n$  is identified with the integer  $\sum_k w_k 2^{k-1} \in [0, N - 1]$ .

## 1 Table encoding and two extraction identities

**Definition 1** (Table encoding, reversal). For a table  $f : \{0, 1\}^n \rightarrow \mathbb{F}$ ,  $V_f(X) = \sum_w f(w)X^w \in \mathbb{F}[X]_{<N}$  and  $\text{Enc}^T(f) = \text{ev}_L(V_f)$ . For  $P \in \mathbb{F}[X]_{<N}$ , its reversal is  $P^*(X) = X^{N-1}P(1/X)$ , so  $[X^j]P^* = [X^{N-1-j}]P$  and  $(P^*)^* = P$ .

$f \mapsto V_f$  is a linear bijection onto  $\mathbb{F}[X]_{<N}$ , so  $\text{Enc}^T$  is a linear bijection onto  $\mathcal{C}_0$  (as KF Lemma 6.3(1)); it is the Kronecker encoding of the multilinear polynomial whose coefficients are the values of  $f$ , and needs no Möbius transform. In particular  $\text{Enc}^T(f) = \text{Enc}(f)$  as maps on vectors of length  $N$ : a table-form commitment is an ordinary Kronecker-FRI commitment to the vector of values, with the same Merkle tree, and all of KF’s results on commitments (unique decoding, extraction, batching) apply to it unchanged.

**Lemma 2** (Evaluation). For  $z \in \mathbb{F}^n$  let  $E_z^*(X) = \prod_{k=1}^n (z_k + (1 - z_k)X^{2^{k-1}})$ . Then  $\tilde{f}(z) = [X^{N-1}]V_f E_z^*$ , where  $\tilde{f}$  is the multilinear extension of  $f$ .

*Proof.* Expanding the product,  $E_z(X) := \sum_w \text{eq}(w, z) X^w = \prod_k ((1 - z_k) + z_k X^{2^{k-1}})$ , and  $X^{N-1} E_z(1/X) = \prod_k X^{2^{k-1}} ((1 - z_k) + z_k X^{-2^{k-1}}) = E_z^*(X)$  since  $N - 1 = \sum_k 2^{k-1}$ . Hence  $[X^{N-1-w}] E_z^* = [X^w] E_z = \text{eq}(w, z)$  and  $[X^{N-1}] V_f E_z^* = \sum_w f(w) \text{eq}(w, z) = \tilde{f}(z)$ .  $\square$

**Lemma 3** (Inner product). *For tables  $a, b$ :  $\sum_w a(w)b(w) = [X^{N-1}] V_a V_b^*$ .*

*Proof.*  $[X^{N-1}] V_a V_b^* = \sum_{i=0}^{N-1} [X^i] V_a [X^{N-1-i}] V_b^* = \sum_i [X^i] V_a [X^i] V_b = \sum_w a(w)b(w)$ .  $\square$

Both are instances of one fact: for  $U \in \mathbb{F}[X]_{<N}$  and a public  $K \in \mathbb{F}[X]_{<N}$ ,  $[X^{N-1}] UK = \sum_i [X^i] U [X^{N-1-i}] K$ . KF's split lemma (Lemma 4.10) and identity lemma (Lemma 4.12) use only  $\deg U, \deg K \leq N - 1$ , so they hold verbatim for  $UK$  replaced by  $V_a E_z^*$  or  $V_a V_b^*$ :

**Lemma 4** (Split and identity, general form). *Let  $U, K, A, H \in \mathbb{F}[X]_{<N}$  and  $v \in \mathbb{F}$ . (1) There are  $A, H \in \mathbb{F}[X]_{<N}$  with  $XUK = A + vX^N + X^{N+1}H$  iff  $v = [X^{N-1}] UK$ . (2)  $\Phi = XUK - A - vX^N - X^{N+1}H$  has degree at most  $2N$ , and  $\Phi = 0$  implies  $v = [X^{N-1}] UK$ .*

*Proof.* As in KF Lemmas 4.10 and 4.12:  $\deg(UK) \leq 2N - 2$ , and the three parts of the right-hand side occupy the disjoint degree ranges  $[0, N - 1]$ ,  $\{N\}$ ,  $[N + 1, 2N]$ ; compare the coefficients of  $X^N$ .  $\square$

**Lemma 5** (Reversed words). *For  $w \in \mathbb{F}^L$  put  $w^*(\xi) = \xi^{N-1} w(\xi^{-1})$ . Then: (1)  $w = \text{ev}_L(P)$  with  $P \in \mathbb{F}[X]_{<N}$  implies  $w^* = \text{ev}_L(P^*)$ ; in particular  $w \mapsto w^*$  maps  $\mathcal{C}_0$  onto  $\mathcal{C}_0$ ; (2)  $w \mapsto w^*$  is a linear bijection of  $\mathbb{F}^L$ , and  $w^* = c^*$  at  $\xi$  iff  $w = c$  at  $\xi^{-1}$ ; (3) a set  $T \subseteq L$  is fibre-closed iff  $T^{-1} = \{\xi^{-1} : \xi \in T\}$  is, and  $|T^{-1}| = |T|$ ; (4)  $h(\xi) = w^*(\xi)$  is computed from the value of  $w$  at  $\xi^{-1} \in L$ , and a fold check at  $\pm\xi$  reads  $w$  on the fibre  $\{\pm\xi^{-1}\}$ .*

*Proof.*  $L$  is a group, so  $\xi \mapsto \xi^{-1}$  is a bijection of  $L$  mapping  $-\xi$  to  $-\xi^{-1}$  (as  $-1 \in L$ ); and  $\xi^{N-1} \neq 0$ . (1)  $\xi^{N-1} P(\xi^{-1}) = P^*(\xi)$ , and  $P \mapsto P^*$  is a bijection of  $\mathbb{F}[X]_{<N}$ . (2)–(4) follow from the first sentence.  $\square$

## 2 The inner-product protocol

**Definition 6** ( $\Pi_{\text{IP}}$ ). Instance: oracles  $w_a, w_b \in \mathbb{F}^L$  and  $S \in \mathbb{F}$ ; honestly  $w_a = \text{Enc}^T(a)$ ,  $w_b = \text{Enc}^T(b)$  and  $S = \sum_w a(w)b(w)$ .

1. The prover sends  $w_A = \text{ev}_L(A)$ , where  $XV_a V_b^* = A + SX^N + X^{N+1}H$  (Lemma 4); the verifier sends  $\beta \in \mathbb{F}$ .
2. Put the virtual words  $w_b^*$  (Lemma 5) and  $h(\xi) = (\xi w_a(\xi) w_b^*(\xi) - w_A(\xi) - S\xi^N)/\xi^{N+1}$ , and

$$w_0 = w_a + \beta w_b^* + \beta^2 w_A + \beta^3 h.$$

3. Run  $\Pi_{\text{FT}}[\ell, \kappa]$  (or  $\Pi_{\text{FT}}^C$ , KF Prop. 5.28) on  $w_0$ .

Per query point the verifier reads  $w_a(\pm\xi)$ ,  $w_b(\pm\xi^{-1})$  and  $w_A(\pm\xi)$ , and does  $O(1)$  extra field operations beyond KF (no kernel to evaluate).

**Theorem 7** (Completeness). *If  $w_a = \text{ev}_L(V_a)$ ,  $w_b = \text{ev}_L(V_b)$  and  $S = \sum_w a(w)b(w)$ , the honest prover is accepted with probability 1.*

*Proof.* By Lemma 3 and Lemma 4(1),  $A, H \in \mathbb{F}[X]_{<N}$  exist; by Lemma 5(1)  $w_b^* = \text{ev}_L(V_b^*)$ ; evaluating the identity at  $\xi$  gives  $h = \text{ev}_L(H)$ . So  $w_0 = \text{ev}_L(V_a + \beta V_b^* + \beta^2 A + \beta^3 H)$  with a polynomial of degree  $< N$ , and KF Lemma 5.9(1) applies.  $\square$

**Definition 8** (Relation).  $\mathcal{R}_{\text{IP}}^\delta$ :  $((w_a, w_b; S), (a, b))$  with  $\Delta_0^{\text{fib}}(w_a, \text{Enc}^T(a)) \leq \delta$ ,  $\Delta_0^{\text{fib}}(w_b, \text{Enc}^T(b)) \leq \delta$  and  $\sum_w a(w)b(w) = S$ .

**Lemma 9** (Batching, degree 3). *Let  $\mathcal{G}$  be the set of  $\beta$  for which  $w_0$  agrees with some  $c \in \mathcal{C}_0$  on a fibre-closed  $T \subseteq L$  with  $|T| \geq (1 - \delta)M$ . If  $|\mathcal{G}| > 3M$ , then  $(w_a, w_b; S)$  has a witness in  $\mathcal{R}_{\text{IP}}^\delta$ .*

*Proof.* The words  $w_0$  form the curve of degree 3 generated by  $(w_a, w_b^*, w_A, h)$ , and every  $\beta \in \mathcal{G}$  has  $\Delta(w_0, \mathcal{C}_0) \leq \delta$ . By correlated agreement for curves (KF Thm 2.16, from [2], with  $t = 3$ ) there are  $P_0, P_1, P_2, P_3 \in \mathbb{F}[X]_{<N}$  and a set  $\text{Agr}$ ,  $|\text{Agr}| \geq (1 - \delta)M$ , on which  $w_a = \text{ev}(P_0)$ ,  $w_b^* = \text{ev}(P_1)$ ,  $w_A = \text{ev}(P_2)$ ,  $h = \text{ev}(P_3)$  (take  $\text{Agr}$  maximal).

*Value.* Let  $\Phi = XP_0P_1 - P_2 - SX^N - X^{N+1}P_3$ . On  $\text{Agr}$ ,  $\Phi(\xi) = \xi w_a(\xi)w_b^*(\xi) - w_A(\xi) - S\xi^N - \xi^{N+1}h(\xi) = 0$  by the definition of  $h$ . Since  $\deg \Phi \leq 2N < (1 - \delta)M \leq |\text{Agr}|$  (KF Lemma 7.2, which uses  $R \geq 2$ ),  $\Phi = 0$ , and Lemma 4(2) gives  $S = [X^{N-1}]P_0P_1$ . Let  $a, b$  be the tables with  $V_a = P_0$  and  $V_b = P_1^*$ ; then  $V_b^* = P_1$  and  $S = \sum_w a(w)b(w)$  by Lemma 3.

*Bad challenges.* For  $\xi \notin \text{Agr}$ ,  $d_\xi(Z) = \sum_{i=0}^3 Z^i(u_i - \text{ev}(P_i))(\xi)$ , with  $(u_0, \dots, u_3) = (w_a, w_b^*, w_A, h)$ , is a nonzero polynomial of degree  $\leq 3$ ; its roots over all such  $\xi$  form a set  $\text{Bad}$  with  $|\text{Bad}| \leq 3\delta M < 3M < |\mathcal{G}|$ . Take  $\beta \in \mathcal{G} \setminus \text{Bad}$  with its  $c$  and  $T$ , and  $\hat{c} = \sum_i \beta^i \text{ev}(P_i) \in \mathcal{C}_0$ . Then  $c$  and  $\hat{c}$  agree with  $w_0$  on  $T \cap \text{Agr}$ , of size  $\geq (1 - 2\delta)M \geq N$ , so  $c = \hat{c}$ ; and for  $\xi \in T$ ,  $d_\xi(\beta) = w_0(\xi) - \hat{c}(\xi) = 0$  forces  $\xi \in \text{Agr}$ . So  $T \subseteq \text{Agr}$ .

*Distances.*  $w_a = \text{ev}(V_a)$  on the fibre-closed  $T$ , so  $\Delta_0^{\text{fib}}(w_a, \text{Enc}^T(a)) \leq \delta$  (KF Lemma 5.11(2)).  $w_b^* = \text{ev}(V_b^*) = \text{ev}(V_b)^*$  on  $T$ , so by Lemma 5(2)  $w_b = \text{ev}(V_b)$  on  $T^{-1}$ , which is fibre-closed of the same size (Lemma 5(3)); so  $\Delta_0^{\text{fib}}(w_b, \text{Enc}^T(b)) \leq \delta$ .  $\square$

**Theorem 10** (Soundness). *If  $(w_a, w_b; S)$  has no witness in  $\mathcal{R}_{\text{IP}}^\delta$ , every prover is accepted with probability at most*

$$\frac{3M}{|\mathbb{F}|} + \varepsilon_{\text{fold}} + (1 - \delta)^\kappa < \frac{4M}{|\mathbb{F}|} + (1 - \delta)^\kappa.$$

*Proof.* Identical to KF Thm 7.8, with Lemma 9 in place of KF Lemma 7.4: for  $\beta \notin \mathcal{G}$  (probability  $\geq 1 - 3M/|\mathbb{F}|$ ) and good  $r$  (probability  $\geq 1 - \varepsilon_{\text{fold}}$ , KF Thm 5.24(1)), the codeword chain (KF Prop. 5.23) shows that  $\text{dens}_\ell(w_\ell) \geq 1 - \delta$  would put  $\beta$  in  $\mathcal{G}$ , the witness set  $\mathcal{W}_\ell(w_\ell)$  being fibre-closed (KF Lemma 7.7); so the query term is at most  $(1 - \delta)^\kappa$  (KF Lemma 3.3).  $\square$

Round-by-round knowledge soundness, the extractor (decode  $w_a$  and  $w_b$  at radius  $\delta^*$ ) and binding follow as in KF Thm 7.11 and Cor. 7.12, with round-1 error  $3M/|\mathbb{F}|$ .

**Remark 11** (One commitment, three kinds of openings). The same table-form commitment  $\text{Enc}^T(f)$  supports: evaluation openings at any  $z$  (KF's protocol with the kernel  $E_z^*$  of Lemma 2, evaluated by the verifier with  $O(n)$  operations); hypercube sums ( $z = (\frac{1}{2}, \dots, \frac{1}{2})$  up to  $2^n$ , or the kernel  $\prod_k (1 + X^{2^{k-1}})$  directly); and inner products with any other committed table ( $\Pi_{\text{IP}}$ ). Inner products with a *public* table  $p$  are the special case  $K = V_p^*$ , useful when  $V_p^*(\xi)$  is cheap to evaluate (product form). All of these batch with KF's batched openings (KF Thm 11.2).

### 3 A sumcheck-free Hadamard check

SNARKs of Spartan/HyperPlonk type need, beyond linear functionals and inner products, the statement  $a(w)b(w) = c(w)$  for all  $w$  (a *Hadamard check*), for three committed tables.

**Geometric weights.** For  $\gamma \in \mathbb{F}$ ,

$$\sum_w \gamma^w a(w)b(w) = [X^{N-1}] V_a(\gamma X) V_b^*(X), \quad \sum_w \gamma^w c(w) = V_c(\gamma),$$

the first by Lemma 3 applied to the tables  $w \mapsto \gamma^w a(w)$  and  $b$ . If  $d = a \circ b - c \neq 0$ , then  $D_d(X) = \sum_w d(w)X^w$  is a nonzero polynomial of degree  $\leq N-1$ , and the two sides differ unless  $D_d(\gamma) = 0$ . The values  $V_a(\gamma\xi)$  and  $V_c(\gamma)$  lie outside  $L$ ; the prover commits to  $Q(X) = V_a(\gamma X)$ , and both are tied to the commitments by DEEP quotients, which are virtual words like  $h$ .

**Definition 12** (Quotient words). For  $w \in \mathbb{F}^L$  and  $y, x \in \mathbb{F}$ , put  $q[w, y, x](\xi) = (w(\xi) - y)/(\xi - x)$ , with the value 0 at  $\xi = x$  if  $x \in L$ .

**Lemma 13** (DEEP step on an agreement set). *Let  $x, y \in \mathbb{F}$ ,  $P, R \in \mathbb{F}[X]_{<N}$  and  $T \subseteq L$  with  $|T| > N+1$ . If  $w = \text{ev}(P)$  and  $q[w, y, x] = \text{ev}(R)$  on  $T$ , then  $P(x) = y$  and  $R = (P - y)/(X - x)$ . Conversely, if  $x \notin L$ ,  $w = \text{ev}_L(P)$  and  $P(x) = y$ , then  $q[w, y, x] = \text{ev}_L((P - y)/(X - x))$ , a polynomial of degree  $\leq N-2$ .*

*Proof.*  $(X - x)R - P + y$  has degree  $\leq N$  and vanishes on  $T \setminus \{x\}$ , which has more than  $N$  points, so it is 0; evaluate at  $x$ . The converse is the factor theorem.  $\square$

**Definition 14** ( $\Pi_{\text{Had}}$ ). Instance: oracles  $w_a, w_b, w_c \in \mathbb{F}^L$ ; honestly  $w_f = \text{Enc}^T(f)$  for  $f \in \{a, b, c\}$  and  $a \circ b = c$ .

1. The verifier sends  $\gamma$ , uniform in  $\mathbb{F}$ . The prover sends  $w_Q = \text{ev}_L(Q)$ , where  $Q(X) = V_a(\gamma X)$ .
2. The verifier sends  $\theta$ , uniform in  $\mathbb{F}$ . The prover sends  $y_1 = V_a(\gamma\theta)$ ,  $y_3 = V_c(\gamma)$  and  $w_A = \text{ev}_L(A)$ , where  $XQV_b^* = A + y_3X^N + X^{N+1}H$  (Lemma 4).
3. The verifier sends  $\beta$ , uniform in  $\mathbb{F}$ . With  $h(\xi) = (\xi w_Q(\xi) w_b^*(\xi) - w_A(\xi) - y_3\xi^N)/\xi^{N+1}$  and

$$(u_0, \dots, u_8) = (w_a, w_c, w_Q, w_b^*, w_A, h, q[w_a, y_1, \gamma\theta], q[w_Q, y_1, \theta], q[w_c, y_3, \gamma]),$$

put  $w_0 = \sum_{i=0}^8 \beta^i u_i$ .

4. Run  $\Pi_{\text{FT}}[\ell, \kappa]$  (or  $\Pi_{\text{FT}}^C$ ) on  $w_0$ .

The challenges are unrestricted: soundness holds for all values, and a pole  $\gamma\theta, \theta, \gamma$  in  $L$  affects only completeness. Using  $y_1$  for both  $V_a(\gamma\theta)$  and  $Q(\theta)$  is the consistency check  $V_a(\gamma\theta) = Q(\theta)$ . Per query point the verifier reads  $w_a, w_c, w_Q, w_A$  on the coset of  $\xi$  and  $w_b$  on the coset of  $\xi^{-1}$ , and does  $O(1)$  field operations and three inversions per point (one batched inversion per query) beyond the folding test.

**Theorem 15** (Completeness). *If  $w_f = \text{Enc}^T(f)$  for  $f \in \{a, b, c\}$  and  $a \circ b = c$ , the honest prover is accepted whenever  $\gamma\theta, \theta, \gamma \notin L$ , hence with probability at least  $1 - 3M/|\mathbb{F}|$ .*

*Proof.*  $Q \in \mathbb{F}[X]_{<N}$ ;  $y_1 = V_a(\gamma\theta) = Q(\theta)$ ; and  $y_3 = V_c(\gamma) = \sum_w \gamma^w c(w) = \sum_w \gamma^w a(w)b(w) = [X^{N-1}] QV_b^*$ , so  $A, H$  exist and  $h = \text{ev}_L(H)$  as in  $\Pi_{\text{IP}}$ . By Lemma 13 the three quotient words are codewords when the poles lie outside  $L$ , and then  $w_0$  is the codeword of a polynomial of degree  $< N$ . Each of the events  $\gamma \in L$ ,  $\theta \in L$ ,  $\gamma\theta \in L$  has probability at most  $M/|\mathbb{F}|$  (for  $\gamma \neq 0$ , at most  $M$  values of  $\theta$  put  $\gamma\theta$  in  $L$ ; for  $\gamma = 0$ , none).  $\square$

**Definition 16** (Relation).  $\mathcal{R}_{\text{Had}}^\delta$ :  $((w_a, w_b, w_c), (a, b, c))$  with  $\Delta_0^{\text{fib}}(w_f, \text{Enc}^T(f)) \leq \delta$  for  $f \in \{a, b, c\}$  and  $a \circ b = c$ .

**Lemma 17** (Batching, degree 8). *Fix  $\gamma, \theta \in \mathbb{F}$  and the prover's  $w_Q, y_1, y_3, w_A$ . Let  $\mathcal{G}$  be the set of  $\beta$  for which  $w_0$  agrees with some  $c' \in \mathcal{C}_0$  on a fibre-closed  $T \subseteq L$  with  $|T| \geq (1 - \delta)M$ . If  $|\mathcal{G}| > 8M$ , there are tables  $a, b, c$  and  $\hat{Q} \in \mathbb{F}[X]_{<N}$  with*

1.  $\Delta_0^{\text{fib}}(w_f, \text{Enc}^T(f)) \leq \delta$  for  $f \in \{a, b, c\}$ , and  $\Delta_0^{\text{fib}}(w_Q, \text{ev}_L(\hat{Q})) \leq \delta$ ;
2.  $V_a(\gamma\theta) = y_1 = \hat{Q}(\theta)$ ;
3.  $V_c(\gamma) = y_3 = [X^{N-1}] \hat{Q} V_b^*$ .

*Proof.* Correlated agreement on one fibre-closed set (the step shared with Lemma 9; Lean: `curve_agreement`,  $t = 8$ ) gives  $P_0, \dots, P_8 \in \mathbb{F}[X]_{<N}$  and a fibre-closed  $T$ ,  $|T| \geq (1 - \delta)M > 2N \geq N + 2$  (KF Lemma 7.2), with  $u_i = \text{ev}(P_i)$  on  $T$ . As in Lemma 9,  $XP_2P_3 - P_4 - y_3X^N - X^{N+1}P_5$  vanishes on  $T$  and has degree  $\leq 2N$ , so  $y_3 = [X^{N-1}] P_2P_3$ . Lemma 13 with  $(P_0, P_6, \gamma\theta)$ ,  $(P_2, P_7, \theta)$  and  $(P_1, P_8, \gamma)$  gives  $P_0(\gamma\theta) = y_1$ ,  $P_2(\theta) = y_1$  and  $P_1(\gamma) = y_3$ . Let  $V_a = P_0$ ,  $V_c = P_1$ ,  $V_b = P_3^*$  and  $\hat{Q} = P_2$ ; then (2) and (3) hold, and (1) follows from agreement on  $T$  (for  $w_b$ , on  $T^{-1}$ , Lemma 5).  $\square$

**Theorem 18** (Soundness). *Let  $R \geq 2$  and  $0 < \delta \leq \delta^*$ . If  $(w_a, w_b, w_c)$  has no witness in  $\mathcal{R}_{\text{Had}}^\delta$ , every prover is accepted with probability at most*

$$\varepsilon_{\text{Had}} = \frac{2(N-1)}{|\mathbb{F}|} + \frac{8M}{|\mathbb{F}|} + \varepsilon_{\text{fold}} + (1 - \delta)^\kappa.$$

*Proof.* For  $\delta \leq \delta^*$  each word has at most one codeword within fibre distance  $\delta$  (KF Lemma 6.3(2)), so the tables and  $\hat{Q}$  of Lemma 17 are the decodings of  $w_a, w_b, w_c, w_Q$ . It suffices to consider deterministic provers (a randomised prover is an average of deterministic ones), so  $w_Q$  is a function of  $\gamma$  and the round-2 messages are functions of  $(\gamma, \theta)$ . Call  $(\gamma, \theta)$  *good* if, for every choice of the prover's round-2 messages,  $|\mathcal{G}| \leq 8M$ . For good  $(\gamma, \theta)$  the argument of KF Thm 7.8 applied to the rest of the protocol bounds the acceptance probability by  $8M/|\mathbb{F}| + \varepsilon_{\text{fold}} + (1 - \delta)^\kappa$ . It remains to bound the probability that  $(\gamma, \theta)$  is not good.

*Case 1:* some  $w_f$  has no table within fibre distance  $\delta$ . Lemma 17(1) fails, so every  $(\gamma, \theta)$  is good.

*Case 2:*  $w_a, w_b, w_c$  decode to  $\hat{a}, \hat{b}, \hat{c}$  with  $\hat{a} \circ \hat{b} \neq \hat{c}$ . Let  $D = D_{\hat{a} \circ \hat{b} - \hat{c}} \neq 0$ . The event  $D(\gamma) = 0$  has probability at most  $(N-1)/|\mathbb{F}|$ . Fix  $\gamma$  with  $D(\gamma) \neq 0$ , and the prover's  $w_Q$  (a function of  $\gamma$ ). If  $w_Q$  has no codeword within fibre distance  $\delta$ ,  $(\gamma, \theta)$  is good for every  $\theta$  by Lemma 17(1). If it decodes to  $\hat{Q} = V_{\hat{a}}(\gamma X)$ , then Lemma 17(3) would give  $\sum_w \gamma^w \hat{c}(w) = V_{\hat{c}}(\gamma) = [X^{N-1}] V_{\hat{a}}(\gamma X) V_{\hat{b}}^* = \sum_w \gamma^w \hat{a}(w) \hat{b}(w)$ , i.e.  $D(\gamma) = 0$ ; so every  $\theta$  is good. Otherwise  $\hat{Q} - V_{\hat{a}}(\gamma X)$  is a nonzero polynomial of degree  $\leq N-1$ , fixed before  $\theta$  is drawn; the event  $\hat{Q}(\theta) = V_{\hat{a}}(\gamma\theta)$  has probability at most  $(N-1)/|\mathbb{F}|$ , and outside it Lemma 17(2) fails, so  $\theta$  is good.  $\square$

So  $\varepsilon_{\text{Had}} < (8M + 2N)/|\mathbb{F}| + \varepsilon_{\text{fold}} + (1 - \delta)^\kappa$ , against  $3M/|\mathbb{F}|$  for  $\Pi_{\text{IP}}$  and  $2M/|\mathbb{F}|$  for an evaluation. The curve has degree 8 because  $w_a$  and  $w_c$  must be in the batch themselves: a quotient word alone determines  $w_a$  on  $T$  only as a polynomial of degree  $\leq N$ , which does not identify the decoding.

Knowledge follows: if the acceptance probability exceeds  $\varepsilon_{\text{Had}}$ , the instance has a witness, and the extractor decodes  $w_a, w_b, w_c$  at radius  $\delta$  (KF Cor. 7.12).

**Remark 19** (Consequence). Linear functionals (evaluations, with the kernel  $E_z^*$ ), inner products ( $\Pi_{\text{IP}}$ ) and Hadamard checks ( $\Pi_{\text{Had}}$ ) are all proved on table-form Kronecker-FRI commitments by one folding test each, with no sumcheck and a verifier that is an FRI verifier plus  $O(1)$  operations per query point (plus  $O(n)$  for  $E_z^*$ ). For R1CS-type SNARKs what remains is the sparse matrix-vector products  $Az, Bz, Cz$ , which Spartan handles with its own sparse-polynomial commitment; whether those admit a coefficient-extraction form is the next question.

## 4 One folding test for a batch of statements

A SNARK needs many statements at once on the same commitments: Hadamard checks, inner products and evaluations. Since every statement of Sections 1–3 is checked by words in one curve, all of them can share one curve and one folding test.

**Definition 20** (Statements). Let  $w_1, \dots, w_r \in \mathbb{F}^L$  be commitments. A *statement* is one of the following, with indices in  $\{1, \dots, r\}$ :

- **Eval** $(i, z, v)$  with  $z \in \mathbb{F}^n$ ,  $v \in \mathbb{F}$ : the table  $f_i$  of  $w_i$  has  $\tilde{f}_i(z) = v$ ;
- **IP** $(i, i', S)$  with  $S \in \mathbb{F}$ :  $\sum_w f_i(w) f_{i'}(w) = S$ ;
- **Had** $(i, i', k)$ :  $f_i \circ f_{i'} = f_k$ .

For a list  $\Sigma = (\sigma_1, \dots, \sigma_J)$  of  $J \geq 1$  statements, let  $D$  be the set of commitment indices read directly ( $i$  in **Eval** and **IP**;  $i, k$  in **Had**),  $B$  the set of indices read reversed ( $i'$  in **IP** and **Had**), and  $\mathcal{H} \subseteq \{1, \dots, J\}$  the positions of the Hadamard checks. A table may play several roles. The relation  $\mathcal{R}_{\text{Batch}}^\delta$  holds for  $((w_i)_i, \Sigma)$  and tables  $(f_i)_{i \in D \cup B}$  if  $\Delta_0^{\text{fib}}(w_i, \text{Enc}^T(f_i)) \leq \delta$  for all  $i \in D \cup B$  and every statement of  $\Sigma$  holds for these tables.

For each statement  $\sigma_j$  the protocol uses a triple  $(U_j, K_j, v_j)$ , with  $U_j, K_j \in \mathbb{F}[X]_{<N}$ , and the identity  $XU_jK_j = A_j + v_jX^N + X^{N+1}H_j$  of Lemma 4:  $(V_{f_i}, E_z^*, v)$  for **Eval** $(i, z, v)$  (Lemma 2);  $(V_{f_i}, V_{f_{i'}}^*, S)$  for **IP** $(i, i', S)$  (Lemma 3);  $(Q_j, V_{f_{i'}}^*, y_3^{(j)})$  with  $Q_j(X) = V_{f_i}(\gamma X)$  for  $\sigma_j = \text{Had}(i, i', k)$  (Section 3). The verifier reads  $U_j(\xi)$  and  $K_j(\xi)$  as  $w_i(\xi)$  and  $E_z^*(\xi)$ , as  $w_i(\xi)$  and  $w_{i'}^*(\xi)$ , or as  $w_Q^{(j)}(\xi)$  and  $w_{i'}^*(\xi)$ .

**Definition 21** ( $\Pi_{\text{Batch}}$ ). Instance:  $(w_i)_i$  and  $\Sigma$ .

1. The verifier sends  $\gamma$ , uniform in  $\mathbb{F}$ . For each  $j \in \mathcal{H}$  the prover sends  $w_Q^{(j)} = \text{ev}_L(Q_j)$ .
2. The verifier sends  $\theta$ , uniform in  $\mathbb{F}$ . For each  $j \in \mathcal{H}$ , with  $\sigma_j = \text{Had}(i, i', k)$ , the prover sends  $y_1^{(j)} = V_{f_i}(\gamma\theta)$  and  $y_3^{(j)} = V_{f_k}(\gamma)$ ; for each  $j \leq J$  it sends  $w_A^{(j)} = \text{ev}_L(A_j)$ .
3. The verifier sends  $\beta$ , uniform in  $\mathbb{F}$ . The words are, in a fixed order:  $w_i$  for  $i \in D$ ;  $w_i^*$  for  $i \in B$ ;  $w_Q^{(j)}$  for  $j \in \mathcal{H}$ ;  $w_A^{(j)}$  and  $h_j(\xi) = (\xi U_j(\xi)K_j(\xi) - w_A^{(j)}(\xi) - v_j\xi^N)/\xi^{N+1}$  for  $j \leq J$ ; and  $q[w_i, y_1^{(j)}, \gamma\theta]$ ,  $q[w_Q^{(j)}, y_1^{(j)}, \theta]$ ,  $q[w_k, y_3^{(j)}, \gamma]$  for  $j \in \mathcal{H}$ ,  $\sigma_j = \text{Had}(i, i', k)$ . They are  $t+1 = |D| + |B| + 2J + 4|\mathcal{H}|$  words  $u_0, \dots, u_t$ ; put  $w_0 = \sum_{s=0}^t \beta^s u_s$ .
4. Run  $\Pi_{\text{FT}}[\ell, \kappa]$  (or  $\Pi_{\text{FT}}^C$ ) on  $w_0$ .

$\gamma$  and  $\theta$  are shared by all Hadamard checks. All  $w_Q^{(j)}$  can be committed in one Merkle tree whose leaves hold one value of each, and likewise all  $w_A^{(j)}$ ; a query then opens, beyond the folding test, one path per commitment in  $D$ , one per commitment in  $B$  (on the inverse coset), and at most two paths for the prover's oracles (one if  $\mathcal{H} = \emptyset$ ), whatever  $J$ .

**Theorem 22** (Completeness). *If  $w_i = \text{Enc}^T(f_i)$  and every statement holds, the honest prover is accepted whenever  $\mathcal{H} = \emptyset$  or  $\gamma\theta, \theta, \gamma \notin L$ ; so with probability at least  $1 - 3M/|\mathbb{F}|$ .*

*Proof.* Each  $w_A^{(j)}$ ,  $h_j$  and quotient word is a codeword, by the completeness proofs of Sections 2 and 3 and Lemma 2 for **Eval** (where  $E_z^*$  is a public polynomial of degree  $< N$ ).  $\square$

**Lemma 23** (Batching, degree  $t$ ). *Fix  $\gamma, \theta$  and the prover's messages, and let  $\mathcal{G}$  be the set of  $\beta$  for which  $w_0$  agrees with some  $c' \in \mathcal{C}_0$  on a fibre-closed  $T \subseteq L$  with  $|T| \geq (1 - \delta)M$ . Let  $R \geq 2$  and  $0 < \delta \leq \delta^*$ . If  $|\mathcal{G}| > tM$ , there are tables  $(f_i)_{i \in D \cup B}$  and polynomials  $\hat{Q}_j \in \mathbb{F}[X]_{<N}$  ( $j \in \mathcal{H}$ ) such that  $\Delta_0^{\text{fib}}(w_i, \text{Enc}^T(f_i)) \leq \delta$ ,  $\Delta_0^{\text{fib}}(w_Q^{(j)}, \text{ev}_L(\hat{Q}_j)) \leq \delta$ , and*

1. *every Eval and IP statement of  $\Sigma$  holds for the  $f_i$ ;*
2. *for every  $j \in \mathcal{H}$  with  $\sigma_j = \text{Had}(i, i', k)$ :  $V_{f_i}(\gamma\theta) = y_1^{(j)} = \hat{Q}_j(\theta)$  and  $V_{f_k}(\gamma) = y_3^{(j)} = [X^{N-1}] \hat{Q}_j V_{f_{i'}}^*$ .*

*Proof.* Correlated agreement on one fibre-closed set (**curve\_agreement** with this  $t$ ) gives polynomials of degree  $< N$  for all the words and a fibre-closed  $T$  with  $|T| \geq (1 - \delta)M > 2N$  on which all

words agree with them. Tables:  $V_{f_i}$  is the polynomial of  $w_i$  for  $i \in D$ ; for  $i \in B$ ,  $V_{f_i}$  is the reversal of the polynomial of  $w_i^*$ , and  $w_i$  agrees with  $\text{Enc}^T(f_i)$  on  $T^{-1}$  (Lemma 5). If  $i \in D \cap B$ , both give tables within fibre distance  $\delta$  of  $w_i$ , which coincide by unique decoding. For each  $j$ , the identity of  $h_j$  on  $T$  and Lemma 4(2) give  $v_j = [X^{N-1}]U_jK_j$  with the polynomials of the words; for **Eval** this is  $v = f_i(z)$  by Lemma 2, for **IP** it is  $S = \sum_w f_i(w)f_{i'}(w)$  by Lemma 3, and for **Had** it is the second equality of (2) with  $\hat{Q}_j$  the polynomial of  $w_Q^{(j)}$ . The three quotient words of each  $j \in \mathcal{H}$  give the other equalities of (2) by Lemma 13, as in Lemma 17.  $\square$

**Theorem 24** (Soundness). *Let  $R \geq 2$  and  $0 < \delta \leq \delta^*$ . If  $((w_i)_i, \Sigma)$  has no witness in  $\mathcal{R}_{\text{Batch}}^\delta$ , every prover is accepted with probability at most*

$$\varepsilon_{\text{Batch}} = \frac{2(N-1)}{|\mathbb{F}|} + \frac{tM}{|\mathbb{F}|} + \varepsilon_{\text{fold}} + (1-\delta)^\kappa.$$

*Proof.* As for Theorem 18: call  $(\gamma, \theta)$  good if  $|\mathcal{G}| \leq tM$  for all round-2 messages; for good  $(\gamma, \theta)$  the acceptance probability is at most  $tM/|\mathbb{F}| + \varepsilon_{\text{fold}} + (1-\delta)^\kappa$ . If some  $w_i$ ,  $i \in D \cup B$ , has no table within fibre distance  $\delta$ , every pair is good by Lemma 23. Otherwise let  $\hat{f}_i$  be the decodings; some statement  $\sigma_{j^*}$  is false for them, and  $j^*$  depends only on the instance. If  $\sigma_{j^*}$  is an **Eval** or **IP** statement, Lemma 23(1) fails, so every pair is good. If  $\sigma_{j^*} = \text{Had}(i, i', k)$ , the proof of Theorem 18 applies verbatim to this one check, with Lemma 23(2) for  $j^*$  in place of Lemma 17: the pair is good unless  $D_{\hat{f}_i \circ \hat{f}_{i'} - \hat{f}_k}(\gamma) = 0$ , or  $\hat{Q}_{j^*} \neq V_{\hat{f}_i}(\gamma X)$  and  $\hat{Q}_{j^*}(\theta) = V_{\hat{f}_i}(\gamma \theta)$ , events of probability at most  $(N-1)/|\mathbb{F}|$  each.  $\square$

When  $\mathcal{H} = \emptyset$  the same proof shows that every pair is good, so the first term can be omitted (this refinement is not in the Lean formalisation). Only one false statement is needed in the proof, so the term  $2(N-1)/|\mathbb{F}|$  from  $\gamma, \theta$  is paid once, not once per Hadamard check; the term  $tM/|\mathbb{F}|$  grows linearly in the number of words, by  $4M/|\mathbb{F}|$  per Hadamard check and  $2M/|\mathbb{F}|$  per other statement. For  $|\mathbb{F}| = p^2 \approx 2^{128}$ ,  $M = 2^{22}$  and  $t = 100$ , this term is about  $2^{-99}$ . Knowledge follows as before: if the acceptance probability exceeds  $\varepsilon_{\text{Batch}}$ , the instance has a witness, obtained by decoding each  $w_i$ .

| $n$ | one batched proof ( $t+1=16$ ) |             |             | four separate proofs |             |             |
|-----|--------------------------------|-------------|-------------|----------------------|-------------|-------------|
|     | prove (ms)                     | verify (ms) | proof (KiB) | prove (ms)           | verify (ms) | proof (KiB) |
| 12  | 38.2                           | 7.21        | 397.9       | 54.3                 | 13.85       | 795.4       |
| 14  | 165.2                          | 8.20        | 479.5       | 224.9                | 16.17       | 1017.2      |
| 16  | 716.2                          | 8.90        | 587.5       | 959.8                | 18.70       | 1315.3      |
| 18  | 3154.6                         | 10.22       | 701.2       | 4116.7               | 22.59       | 1658.7      |
| 20  | 13187.4                        | 11.32       | 804.5       | 18012.6              | 25.43       | 1957.9      |

Table 1: The statements **Had**( $a, b, c$ ), **IP**( $a, d, S$ ), **Eval**( $a, z_1, v_1$ ), **Eval**( $c, z_2, v_2$ ) on four table-form commitments, proved by one  $\Pi_{\text{Batch}}$  proof or by four (each a  $\Pi_{\text{Batch}}$  proof of one statement). Configuration KF-8, 148 queries, 32-byte salts,  $\mathbb{F}_{p^2}$ ; one core; medians. Source: `bench/batch.csv`, `cargo run --release --example batch_bench`. The prototype prover is not optimised.

One proof for the four statements is 2.0 to 2.4 times smaller and 1.9 to 2.2 times faster to verify than four proofs (2.4 and 2.2 at  $n = 20$ ), and its prover takes 23 to 30% less time.

## 5 Sparse matrix–vector products without sumcheck

An R1CS instance asks for  $z$  with  $(Az) \circ (Bz) = Cz$  for public sparse matrices  $A, B, C$ . Committing to  $z$  and to  $a = Az$ ,  $b = Bz$ ,  $c = Cz$ , the statement  $a \circ b = c$  is a Hadamard check. This section

proves  $a = \mathbf{M}z$  for a public sparse matrix  $\mathbf{M}$  (a *lincheck*) with the statements of Section 4 on affine forms of committed and public tables; lookups are handled with logarithmic derivatives (LogUp, Haböck [4]). Throughout,  $N$  is at least the length of  $z$  and the number of nonzero entries of each matrix, and all tables are padded to length  $N = 2^n$ ;  $R \geq 2$  and  $0 < \delta \leq \delta^*$ . Several matrices  $\mathbf{M}_1, \dots, \mathbf{M}_J$  are handled together (for R1CS,  $J = 3$ ), and  $\text{char } \mathbb{F} > JN$ .

## 5.1 Affine forms and public tables

A *public table* is a table  $P$  whose polynomial  $V_P$  (and hence  $V_P^*$ ) the verifier evaluates at any point with few operations. We use

$$\mathbf{0}(w) = 1, \quad \mathbf{0}(w) = 0, \quad \text{id}(w) = w, \quad G_\eta(w) = \eta^w \quad (0^0 = 1),$$

with  $w$  read as an integer in  $[0, N - 1]$  and then as an element of  $\mathbb{F}$ , for which  $V_1(X) = \prod_{k < n} (1 + X^{2^k}) = V_1^*(X)$ ,  $V_{G_\eta}(X) = V_1(\eta X)$ ,  $V_{G_\eta}^*(X) = \prod_{k < n} (X^{2^k} + \eta^{2^k})$ , and  $V_{\text{id}} = T_n$  with  $S_0 = 1$ ,  $T_0 = 0$ ,  $S_{k+1} = (1 + X^{2^k})S_k$ ,  $T_{k+1} = (1 + X^{2^k})T_k + 2^k X^{2^k} S_k$ ; each costs  $O(n)$  operations. An *affine form* is  $\lambda = \sum_i c_i w_i + \sum_j d_j P_j$  with commitments  $w_i$ , public tables  $P_j$  and coefficients  $c_i, d_j \in \mathbb{F}$ . Its word at  $\xi$  is  $\sum_i c_i w_i(\xi) + \sum_j d_j V_{P_j}(\xi)$ , and for tables  $f_i$  of the  $w_i$  its table is  $\sum_i c_i f_i + \sum_j d_j P_j$ .

**Proposition 25** ( $\Pi_{\text{Batch}}$  on affine forms). *Let the arguments of the statements of Section 4 be affine forms, the coefficients and the parameters of the public tables being part of the instance.*

- *Relation: tables are required for the committed words that occur with a nonzero coefficient in some argument, within fibre distance  $\delta$ ; every statement holds for the tables of its affine forms. Public tables carry no condition.*
- *Curve: the committed words occurring directly, the reversals of those occurring reversed, every  $w_Q^{(j)} = \text{ev}_L(V_\lambda(\gamma X))$  (public part included), every  $w_A^{(j)}$  and  $h_j$ , and the quotient words  $q[\lambda, y, x]$  of affine forms  $\lambda$  with a committed part. When the form of a quotient word is purely public, the verifier checks  $y = V_\lambda(x)$  directly instead.*

*Then completeness, Lemma 23 and Theorem 24 hold with  $t + 1$  the number of words in the curve. The verifier's work per query point grows by  $O(n)$  per public table.*

*Proof.* On the agreement set  $T$  of Lemma 23, the word of an affine form agrees with the polynomial  $\sum_i c_i P_{w_i} + \sum_j d_j V_{P_j}$  of degree  $< N$ , since public tables agree with their polynomials everywhere. The proofs of Lemma 23 and Theorem 24 use only such agreements, Lemmas 4 and 13, and the decodings of the committed words, which determine the tables of the affine forms; the case analysis of Theorem 18 uses only the polynomial  $V_\lambda$  of the decoded form. A direct check  $y = V_\lambda(x)$  gives the conclusion of Lemma 13 for a public form.  $\square$

## 5.2 The lincheck

**Definition 26** (Index, relation). A sparse  $\mathbf{M}_j \in \mathbb{F}^{N \times N}$  ( $j < J$ ) with at most  $N$  nonzero entries is listed as  $(r_{j,k}, c_{j,k}, v_{j,k})_{k < N}$ , padded with  $v_{j,k} = 0$ ,  $r_{j,k} = c_{j,k} = 0$ . The *index* of  $(\mathbf{M}_j)_{j < J}$  consists of the commitments  $\text{Enc}^T$  of the tables  $\text{row}_j(k) = r_{j,k}$ ,  $\text{col}_j(k) = c_{j,k}$ ,  $\text{val}_j(k) = v_{j,k}$  and of the *combined* multiplicities  $m_R(w) = \#\{(j, k) : r_{j,k} = w\}$ ,  $m_C(w) = \#\{(j, k) : c_{j,k} = w\}$ , computed once by an honest indexer (soundness uses only that the  $\text{row}_j, \text{col}_j, \text{val}_j$  and  $m_R, m_C$  are honest), all in one Merkle tree. The lincheck relation holds for commitments  $(w_z, (w_{a_j})_j)$  and tables  $(z, (a_j)_j)$  if all are within fibre distance  $\delta$  of  $\text{Enc}^T$  of the tables and  $a_j = \mathbf{M}_j z$  for every  $j$ .

The reduction rests on  $\sum_w \eta^w (\mathbf{M}_j z)(w) = \sum_k v_{j,k} \eta^{r_{j,k}} z(c_{j,k})$ : the prover commits to  $e_{j,k} = \eta^{r_{j,k}}$ ,  $\zeta_{j,k} = z(c_{j,k})$  and  $p_{j,k} = e_{j,k} \zeta_{j,k}$ , and proves the  $JN$  lookups  $(r_{j,k}, e_{j,k}) \in \{(w, \eta^w)\}$  with *one* LogUp, and the  $JN$  lookups  $(c_{j,k}, \zeta_{j,k}) \in \{(w, z(w))\}$  with another.

**Definition 27** ( $\Pi_{\text{Lin}}$ ). Instance: the index of  $(\mathbf{M}_j)_{j < J}$  and commitments  $w_z, (w_{a_j})_j$ .

1. The verifier sends  $\eta$ , uniform in  $\mathbb{F}$ . The prover commits to tables  $e_j, \zeta_j, p_j$  (honestly  $e_{j,k} = \eta^{r_{j,k}}, \zeta_{j,k} = z(c_{j,k}), p_{j,k} = e_{j,k} \zeta_{j,k}$ ).
2. The verifier sends  $x_R, y_R, x_C, y_C$ , uniform in  $\mathbb{F}$ . The prover commits to tables  $\phi_j^R, \psi_j^R, \phi_j^C, \psi_j^C$  and sends  $s_j \in \mathbb{F}$ ; honestly

$$\begin{aligned} \phi_{j,k}^R &= \frac{1}{x_R - r_{j,k} - y_R e_{j,k}}, & \psi_w^R &= \frac{m_R(w)}{x_R - w - y_R \eta^w}, \\ \phi_{j,k}^C &= \frac{1}{x_C - c_{j,k} - y_C \zeta_{j,k}}, & \psi_w^C &= \frac{m_C(w)}{x_C - w - y_C z(w)}, \end{aligned}$$

with  $\psi_w = 0$  when  $m(w) = 0$ ; the honest prover aborts if a denominator with  $m(w) \neq 0$  vanishes.

3. The statements, proved by one  $\Pi_{\text{Batch}}$  on affine forms (Proposition 25) whose own challenges  $\gamma, \theta, \beta$  are drawn after step 2: for every  $j$ ,

$$\begin{aligned} &\text{IP}(a_j, G_\eta, s_j), \quad \text{IP}(\text{val}_j, p_j, s_j), \quad \text{Had}(e_j, \zeta_j, p_j), \\ &\text{Had}(\phi_j^R, x_R \mathbf{1} - \text{row}_j - y_R e_j, \mathbf{1}), \quad \text{Had}(\phi_j^C, x_C \mathbf{1} - \text{col}_j - y_C \zeta_j, \mathbf{1}), \end{aligned}$$

and once

$$\begin{aligned} &\text{Had}(\psi^R, x_R \mathbf{1} - \text{id} - y_R G_\eta, m_R), \quad \text{IP}(\sum_j \phi_j^R - \psi^R, \mathbf{1}, 0), \\ &\text{Had}(\psi^C, x_C \mathbf{1} - \text{id} - y_C z, m_C), \quad \text{IP}(\sum_j \phi_j^C - \psi^C, \mathbf{1}, 0), \end{aligned}$$

that is  $5J + 4$  statements.

The honest prover aborts with probability at most  $2N/|\mathbb{F}|$  (a denominator  $x - w - y \cdot$  with  $m(w) \neq 0$  vanishes; the denominators of the  $\phi$  are among them), and is otherwise accepted with the probability of Proposition 25, at least  $1 - 3M/|\mathbb{F}|$ .

**Lemma 28** (LogUp with pairs). *Let pairs  $(\alpha_k, \alpha'_k)_{k < K}$  and  $(\omega_w, \omega'_w)_{w < N_t}$  in  $\mathbb{F}^2$ ,  $m_w \in \mathbb{F}$ , and  $\text{char } \mathbb{F} > K$ . If some  $(\alpha_{k_0}, \alpha'_{k_0})$  is not among the  $(\omega_w, \omega'_w)$ , then for  $x, y$  uniform in  $\mathbb{F}$ ,*

$$\Pr \left[ \sum_k \frac{1}{x - \alpha_k - y \alpha'_k} = \sum_w \frac{m_w}{x - \omega_w - y \omega'_w} \text{ with all denominators nonzero} \right] \leq \frac{K + 2N_t - 1}{|\mathbb{F}|}.$$

*Proof.* Fix  $y$  and write  $c_k = \alpha_k + y \alpha'_k$ ,  $t_w = \omega_w + y \omega'_w$ . The equation  $c_{k_0} = t_w$  is  $\alpha_{k_0} - \omega_w = y(\omega'_w - \alpha'_{k_0})$ ; it has no solution  $y$  when  $\alpha'_{k_0} = \omega'_w$  (then  $\alpha_{k_0} \neq \omega_w$ ) and one otherwise, so  $c_{k_0}$  is a table value  $t_w$  for at most  $N_t$  values of  $y$ . For the other  $y$ , let  $S = \{c_k\}$ ,  $n_c = \#\{k : c_k = c\}$  and

$$\text{Num}(X) = \sum_{c \in S} n_c \prod_{c' \in S \setminus \{c\}} (X - c') \prod_w (X - t_w) - \sum_w m_w \prod_{c \in S} (X - c) \prod_{v \neq w} (X - t_v),$$

of degree  $\leq K + N_t - 1$ . At  $X = c_{k_0}$  only the term of  $c = c_{k_0}$  survives, and it is  $n_{c_{k_0}} \in [1, K]$  (nonzero in  $\mathbb{F}$ ) times nonzero products, so  $\text{Num} \neq 0$ . Where all denominators are nonzero, the equation multiplied by  $\prod_{c \in S} (x - c) \prod_w (x - t_w)$  is  $\text{Num}(x) = 0$ , which holds for at most  $K + N_t - 1$  values of  $x$ .  $\square$

**Theorem 29** (Lincheck reduction). *Suppose the index is honest, the decodings  $z, a_j$  of  $w_z, w_{a_j}$  exist and  $a_{j_0} \neq \mathbf{M}_{j_0} z$  for some  $j_0$ , and fix a deterministic prover. Let  $E$  be the event, over  $\eta, x_R, y_R, x_C, y_C$ , that every word committed by the prover in steps 1–2 has a decoding within fibre distance  $\delta$  and that these decodings, with  $z, a_j$  and the index, satisfy all the statements of step 3. Then*

$$\Pr[E] \leq \varepsilon_{\text{Lin}} = \frac{N - 1}{|\mathbb{F}|} + 2 \frac{N + (JN + 2N - 1)}{|\mathbb{F}|} = \frac{(2J + 7)N - 3}{|\mathbb{F}|}.$$

*Proof.* On  $E$ : from  $\text{Had}(\phi_j^R, x_R \mathbf{1} - \text{row}_j - y_R e_j, \mathbf{1})$ , every  $x_R - r_{j,k} - y_R e_{j,k}$  is nonzero and  $\phi_{j,k}^R$  is its inverse. From  $\text{Had}(\psi^R, \dots, m_R)$ ,  $\psi_w^R(x_R - w - y_R \eta^w) = m_R(w)$ ; unless  $x_R = w + y_R \eta^w$  for some  $w$  (probability  $\leq N/|\mathbb{F}|$  over  $x_R$ , for every  $y_R$ ),  $\psi_w^R = m_R(w)/(x_R - w - y_R \eta^w)$ . The last row statement gives  $\sum_{j,k} \phi_{j,k}^R = \sum_w \psi_w^R$ . The decodings  $e_j$  are fixed before  $x_R, y_R$ , and  $w \mapsto w \in \mathbb{F}$  is injective on  $[0, N-1]$  since  $\text{char } \mathbb{F} > N$ ; so by Lemma 28 with  $K = JN$  and  $N_t = N$ , except with probability  $(JN + 2N - 1)/|\mathbb{F}|$ , every  $(r_{j,k}, e_{j,k})$  is a pair  $(w, \eta^w)$ , that is  $e_{j,k} = \eta^{r_{j,k}}$ . In the same way, except with probability  $(N + JN + 2N - 1)/|\mathbb{F}|$ ,  $\zeta_{j,k} = z(c_{j,k})$  for all  $j, k$  (the decodings  $\zeta_j$  and  $z$  are fixed before  $x_C, y_C$ ). Then  $\text{Had}(e_{j_0}, \zeta_{j_0}, p_{j_0})$  gives  $p_{j_0,k} = \eta^{r_{j_0,k}} z(c_{j_0,k})$ , and the first two statements give  $\sum_w \eta^w a_{j_0}(w) = s_{j_0} = \sum_k v_{j_0,k} \eta^{r_{j_0,k}} z(c_{j_0,k}) = \sum_w \eta^w (\mathbf{M}_{j_0} z)(w)$ . So  $\eta$  is a root of the nonzero polynomial  $\sum_w (a_{j_0} - \mathbf{M}_{j_0} z)(w) X^w$  of degree  $\leq N - 1$ , fixed before  $\eta$ : probability  $\leq (N - 1)/|\mathbb{F}|$ .  $\square$

For  $J = 1$ ,  $\varepsilon_{\text{Lin}} = (9N - 3)/|\mathbb{F}|$ . Separate lookups for each matrix would cost  $J(4N - 1)/|\mathbb{F}|$  per kind of lookup instead of  $((J + 3)N - 1)/|\mathbb{F}|$ , and  $J - 1$  more words  $\psi$  and more statements per kind.

**Theorem 30** (Soundness of the lincheck). *If the index is honest and  $(w_z, (w_{a_j})_j)$  has no witness for the lincheck relation, every prover of  $\Pi_{\text{Lin}}$  followed by  $\Pi_{\text{Batch}}$  is accepted with probability at most  $\varepsilon_{\text{Lin}} + \varepsilon_{\text{Batch}}$ , with  $\varepsilon_{\text{Batch}}$  of Theorem 24 for the  $t + 1$  words of step 3.*

*Proof.* It suffices to consider deterministic provers. Let  $\tau$  be the transcript of steps 1–2; the batch instance  $I(\tau)$  and the event  $E$  of Theorem 29 are functions of  $\tau$ , and given  $\tau$  the rest of the prover is a prover of  $\Pi_{\text{Batch}}$  facing fresh challenges. If  $w_z$  or some  $w_{a_j}$  has no decoding,  $I(\tau)$  has no witness for every  $\tau$ . Otherwise  $a_{j_0} \neq \mathbf{M}_{j_0} z$  for some  $j_0$ , and for  $\tau \notin E$ ,  $I(\tau)$  has no witness: by unique decoding a witness would consist of the decodings, which either do not exist or violate a statement. In both cases Theorem 24 gives  $\Pr[\text{accept} \mid \tau] \leq \varepsilon_{\text{Batch}}$  for  $\tau \notin E$ , so the acceptance probability is at most  $\Pr[E] + \varepsilon_{\text{Batch}}$ .  $\square$

### 5.3 R1CS

**Definition 31** (R1CS relation). Public: sparse  $A, B, C \in \mathbb{F}^{N \times N}$  (with an honest index), input positions  $I \subseteq [0, N - 1]$  and an input  $x \in \mathbb{F}^I$ . Let  $\hat{x}$  be the table equal to  $x$  on  $I$  and 0 elsewhere, and  $\chi$  the 0/1 table of  $I$  (both public tables; for  $I = [0, |I| - 1]$ ,  $V_\chi(X) = (X^{|I|} - 1)/(X - 1)$ , and in general  $V_{\hat{x}}, V_\chi$  cost  $O(|I|)$ ). The relation holds for commitments  $(w_w, w_a, w_b, w_c)$  and tables  $(w, a, b, c)$  if all four are within fibre distance  $\delta$ ,  $w$  vanishes on  $I$ , and with  $z = \hat{x} + w$ :  $a = Az$ ,  $b = Bz$ ,  $c = Cz$  and  $a \circ b = c$ .

**Theorem 32** (R1CS without sumcheck). *Run  $\Pi_{\text{Lin}}$  for  $(\mathbf{M}_0, \mathbf{M}_1, \mathbf{M}_2) = (A, B, C)$  and  $(a_0, a_1, a_2) = (a, b, c)$  (so  $J = 3$  and  $\text{char } \mathbb{F} > 3N$ ) on the affine form  $z = \hat{x} + w_w$ , and prove its 19 statements,  $\text{Had}(a, b, c)$  and  $\text{Had}(w, \chi, \mathbf{0})$  in one  $\Pi_{\text{Batch}}$  on affine forms. If the instance has no witness, every prover is accepted with probability at most*

$$\varepsilon_{\text{Lin}} + \varepsilon_{\text{Batch}} \leq \frac{15N - 5 + tM}{|\mathbb{F}|} + \varepsilon_{\text{fold}} + (1 - \delta)^\kappa.$$

*Proof.* As in Theorem 30, with one false statement: if some decoding is missing, the batch instance never has a witness; if  $a \neq Az$ ,  $b \neq Bz$  or  $c \neq Cz$  for the decodings, only the event  $E$  of the lincheck matters, of probability  $\leq \varepsilon_{\text{Lin}} = (13N - 3)/|\mathbb{F}|$ ; otherwise  $a \circ b \neq c$  or  $w$  does not vanish on  $I$ , and  $\text{Had}(a, b, c)$  or  $\text{Had}(w, \chi, \mathbf{0})$  is false for every value of the challenges.  $\square$

Every step is a coefficient extraction, a lookup by logarithmic derivatives, or a Hadamard check: an R1CS argument on Kronecker-FRI commitments needs no sumcheck.

**Status.** The algebra of  $\Pi_{\text{Lin}}$  is checked in the clear by the Rust module `lincheck` (closed forms of  $V_1$ ,  $V_{\text{id}}$ ,  $V_{G_\eta}$ ; honest runs with padded matrices; a wrong  $a$  caught by the weighted sums; a forged  $\zeta_k$  or  $e_k$  that repairs the weighted sums caught by the column or row lookup). The compiled protocol is implemented: the module `affine` runs  $\Pi_{\text{Batch}}$  on affine forms (Proposition 25) with committed words grouped in Merkle trees whose leaves hold one value of each word of a group, and the module `rics` implements the indexer, prover and verifier of Theorem 32: one index tree of 11 words ( $\text{row}_j, \text{col}_j, \text{val}_j, m_R, m_C$ ), combined row and column lookups, 21 statements and  $t + 1 = 127$  words; the proof opens six trees per query (witness, index, the two `lincheck` groups, the two groups of  $\Pi_{\text{Batch}}$ ). Its tests accept honest proofs and reject a wrong public input, an unsatisfied constraint, a witness that is nonzero on an input position, a false `lincheck` alone, and a false `lincheck` repaired by a forged  $\zeta_k$  or  $e_k$ ; removing the sum statement of the column (row) lookup makes the forged  $\zeta_k$  ( $e_k$ ) accepted, so each lookup is exercised. The Lean formalisation of this section is described in Section 6.

| $n$ | constraints | index (ms) | prove (ms) | verify (ms) | proof (KiB) |
|-----|-------------|------------|------------|-------------|-------------|
| 10  | 512         | 3.1        | 58.4       | 20.5        | 890.0       |
| 12  | 2048        | 12.7       | 265.1      | 40.2        | 1701.1      |
| 14  | 8192        | 53.0       | 1176.4     | 42.0        | 1843.8      |
| 16  | 32768       | 267.7      | 5619.1     | 44.4        | 1997.1      |

Table 2: The R1CS prototype on the instances of `rics::sample_instance` ( $N/2$  constraints  $(Az)_i(Bz)_i = z_{N/2+i}$ , two nonzero entries per row of  $A$  and  $B$ , two public inputs);  $t + 1 = 127$  words; configuration KF-8 ( $\ell = \max(n - 8, 1)$ ), 148 queries, 32-byte salts,  $\mathbb{F}_{p^2}$ ; one core; medians. Source: `bench/rics.csv`, `cargo run --release --example rics_bench`.

Compared with separate lookups for each matrix and three index trees (version 8 of this note: 179 words, eight trees), at  $n = 16$  the prover is 28% faster ( $7802 \rightarrow 5619$  ms), the verifier 35% faster ( $67.8 \rightarrow 44.4$  ms) and the proof 26% smaller ( $2694 \rightarrow 1997$  KiB).

## 6 Relation to known techniques and status

Writing an inner product as the middle coefficient of a product with a reversed polynomial is classical, and appears in pairing-based multilinear commitments (Mercury [3] uses the table form with KZG). DEEP quotients for out-of-domain values are standard in FRI-based STARKs. The new points here are: (1) the reversed virtual word  $w_b^*$ , which reads the second vector from the *same* Reed–Solomon commitment thanks to  $L^{-1} = L$ ; (2) the reuse of Kronecker-FRI’s opening polynomial, virtual word and single folding test, with a fully proved soundness bound (Lemma 9); (3) the table-form evaluation kernel  $E_z^*$ , which gives evaluation openings on the same commitments; (4) a Hadamard check in which the out-of-domain values are DEEP quotient words inside the same batch as the opening polynomial, so that one degree-8 curve and one folding test carry the whole argument (Lemma 17).

**Prover cost and the sumcheck baseline.** The prover of  $\Pi_{\text{IP}}$  computes one product  $V_a V_b^*$  of two polynomials of degree  $< N$  over the base field (three NTTs of size  $2N$ ), one NTT of size  $M$  and one Merkle tree for  $w_A$ , and then runs the folding test; every round-1 object lies in the base field, since  $V_a, V_b$  do. The natural baseline is the sumcheck route:  $n$  rounds of a degree-2 sumcheck on  $\sum_w a(w)b(w)$ , followed by openings of  $a$  and  $b$  at the random point. For that baseline the fastest prover we know of in our own work is the tree circuit of [5], which produces the evaluations,

hypercube sums and sumcheck round polynomials of a multilinear polynomial from one binary-tree recursion; the comparison of the two routes at equal soundness is left to the implementation paper.

### Status.

- *Lean 4.* Sections 1–2 are formalised in the Kronecker-FRI Lean project, files `TableForm.lean` and `InnerProduct.lean`, without `sorry` and with the single axiom of that project (correlated agreement for curves, KF Thm 2.16, here with  $t = 3$ ). Lemma 2: `eval_table`; Lemma 3: `inner_product`; Lemma 4: `splitG_iff`, `natDegree_PhiG_1e`, `identityG`; Lemma 5: `wrev_ev`, `wrev_eq_iff`, `inv_image_closed`, `ncard_inv_image`; completeness: `honestIP_prob_one`; Lemma 9: `batching_IP`; soundness: `soundness_IP`, and `soundnessC_IP` for any folding arity  $2^k$ ; knowledge and binding: `knowledge_IP`, `binding_IP`. The common step of KF Lemma 7.4 and Lemma 9, correlated agreement on one fibre-closed set for a curve of any degree, is the separate theorem `curve_agreement`.
- *Rust.* The module `ip` of the Kronecker-FRI crate implements  $\Pi_{IP}$  with Merkle caps, arity  $2^k$  and Fiat–Shamir, on the commitments of `pcs`. Its tests check Lemmas 3, 4 and 5(1), completeness for  $n \leq 10$ , two rates and arities 2 to 16, and the rejection of three cheating provers (a wrong  $S$ ; a wrong  $S$  absorbed into a polynomial  $A$  of degree  $N$ , as in KF Remark 4.13; opening polynomials computed from a table  $b' \neq b$ ).
- *Lean 4, Section 3.* File `Hadamard.lean`, same axiom, no `sorry`. Lemma 13: `deep_step`, `deep_honest`; completeness: `honestHad_accepts`, and `honestHad_prob` for the bound  $1 - 3M/|\mathbb{F}|$ ; Lemma 17: `batching_Had`; Theorem 18: `prob_bad_1e` (the bad pairs  $(\gamma, \theta)$ ), `soundness_Had`, and `soundnessC_Had` for any folding arity  $2^k$ ; knowledge: `knowledge_Had`. The step “for good  $(\gamma, \theta)$ , the argument of KF Thm 7.8” is the separate theorem `ft_family_bound`: for any family of words  $w_0(\beta)$ , the acceptance probability is at most  $|\mathcal{G}|/|\mathbb{F}| + \varepsilon_{\text{fold}} + (1 - \delta)^\kappa$ .
- *Lean 4, Section 4.* File `BatchStmts.lean`, same axiom, no `sorry`. Statements: `BStmt` (`eval`, `ip`, `had`), `Dset`, `Bset`, `RelBatchS`; the words are indexed by the finite type `BWord` ( $D \oplus B \oplus \mathcal{H} \oplus [J] \oplus [J] \oplus \mathcal{H} \times [3]$ ) with a fixed enumeration, and `curve_agreement_idx` is correlated agreement for such an index type. Lemma 23: `batching_B`; Theorem 24: `prob_bad_had` (one false Hadamard check), `prob_bad_B`, `soundness_B`, `soundnessC_B` (any folding arity); knowledge: `knowledge_B`; completeness: `bwords_honest`, `honestB_accepts`, `honestB_prob`.
- *Lean 4, Section 5.* File `Lincheck.lean`, no `sorry`, and no axiom beyond the three standard ones of Lean and Mathlib. Lemma 28: `logup_prob`, for any finite index types of the  $K$  lookups and  $N_t$  table pairs (numerator `logupNum`, `logupNum_ne_zero`, `logupNum_eval_eq_zero`). Theorem 29: `lincheck_reduction`, for  $J$  matrices with combined lookups, at the level of tables: any deterministic prover given by its tables as functions of the challenges (`LinProver`), the statements of step 3 on these tables (`HoldsLin`), and the bound  $(N - 1 + 2(N + (N + (JN + N - 1))))/|\mathbb{F}|$ ; one failing lookup: `prob_lookupFail`. Theorems 30 and 32: `lincheck_sound` and `r1cs_sound`, via the two-phase bound `prob_two_phase`, with the second phase abstract: they hold for any second phase whose acceptance probability is  $\leq \varepsilon_2$  whenever a statement of step 3 fails (resp. a statement of step 3, `Had(a, b, c)` or `Had(w,  $\chi$ , 0)`). Not formalised: Proposition 25, i.e. that  $\Pi_{\text{Batch}}$  on affine forms with public tables provides that second phase with  $\varepsilon_2 = \varepsilon_{\text{Batch}}$  (`soundness_B` is proved for statements on committed words only), and the closed forms of  $V_1$ ,  $V_{\text{id}}$ ,  $V_{G_\eta}$  (checked by the Rust tests).
- *Rust, Section 4.* The module `batch` implements  $\Pi_{\text{Batch}}$  with one multi-value Merkle tree for all  $w_Q$  and one for all  $w_A$ ; its tests check Lemma 2 for  $E_z^*$ , completeness for mixed batches (two Hadamard checks sharing a table, an inner product, two evaluations) for  $n \leq 10$  and arities 2 to 16, and the rejection of a batch with one false evaluation, one false inner product,

or one false Hadamard check (also against a prover that sets  $y_3$  to the weighted inner product, which only the quotient word  $q[w_c, y_3, \gamma]$  catches).

- *Rust, Section 3.* The module `had` implements  $\Pi_{\text{Had}}$  on the shared folding-test code of module `ft`, drawing  $\gamma, \theta$  by rejection sampling outside the base field  $\mathbb{F}_p \supseteq L$ , so that no pole lies in  $L$  and completeness is perfect (the two first terms of  $\varepsilon_{\text{Had}}$  become  $(N - 1)/(|\mathbb{F}| - p)$  each); its tests check completeness for  $n \leq 10$ , two rates and arities 2 to 16, and the rejection of a wrong  $y_1$ , of a wrong  $Q$  (with  $y_1 = Q(\theta)$ ), and of a false instance ( $c$  differing from  $a \circ b$  in one entry) with three choices of  $y_3$ . Removing  $q[w_c, y_3, \gamma]$  from the batch makes the false instance accepted, and a wrong pole in the verifier breaks completeness, so the tests exercise each part.

| $n$ | prove (ms) |        |        | verify (ms) |      |      | proof (KiB) |       |       |
|-----|------------|--------|--------|-------------|------|------|-------------|-------|-------|
|     | KF         | IP     | Had    | KF          | IP   | Had  | KF          | IP    | Had   |
| 12  | 7.5        | 6.2    | 17.6   | 2.09        | 2.13 | 4.31 | 138.1       | 171.4 | 289.1 |
| 14  | 32.2       | 25.9   | 74.0   | 2.56        | 2.60 | 4.97 | 182.3       | 233.4 | 371.1 |
| 16  | 141.4      | 109.2  | 314.7  | 3.24        | 3.25 | 5.63 | 249.8       | 308.1 | 458.8 |
| 18  | 548.2      | 437.8  | 1353.6 | 3.87        | 4.01 | 6.40 | 325.0       | 395.3 | 565.7 |
| 20  | 2736.6     | 1947.4 | 6004.5 | 4.59        | 5.13 | 7.39 | 391.7       | 471.1 | 662.0 |

Table 3: KF: one evaluation opening ( $\Pi_{\text{KF}}$ ); IP: one inner product ( $\Pi_{\text{IP}}$ ); Had: one Hadamard check ( $\Pi_{\text{Had}}$ ). Configuration KF-8 of the Kronecker-FRI paper (rate 1/4, arity 8, caps of 128 nodes,  $\ell = n - 8$ ), 148 queries, 32-byte salts,  $\mathbb{F}_{p^2}$  over Goldilocks; one core of an Intel Xeon at 2.10 GHz; medians; commitments not included. Source: `bench/ip.csv`, `cargo run --release --example ip_bench`.

An inner product of two committed tables costs less prover time than one evaluation opening (the round-1 work is in the base field), about the same verifier time, and about 20% more proof, the extra openings of  $w_b$  on the inverse cosets. A Hadamard check costs about 2.2 evaluation openings of prover time (its polynomials  $Q, A, H$  lie in the extension field) and 1.7 times the proof, since five oracles are opened per query instead of two; when  $a, b, c$  are committed together, one tree with three values per leaf removes two of the five authentication paths per query.

**Next.** The Lean formalisation of Proposition 25 (the batch on affine forms and public tables), which would close the R1CS theorem end to end; a comparison with a sumcheck-based R1CS argument (Spartan-style) at equal soundness.

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